IMAGE ENHANCEMENT

Using math operation and statistics to adjust pixel intensity

PIXEL BASED OPERATION

Adjust Pixel Intensity

PIXEL-BASED OPERATION

WHAT WOULD BE THE EFFECTS OF CHANGING PIXEL INTENSITY?

http://what-when-how.com/introduction-to-video-and-image-processing/visual-effects-introduction-to-video-and-image-processing-part-1/

http://www.cs.uregina.ca/Links/class-info/425/Lab3/

IF WE PERFORM

MATH OPERATION

ON AN IMAGE,

What will happen?

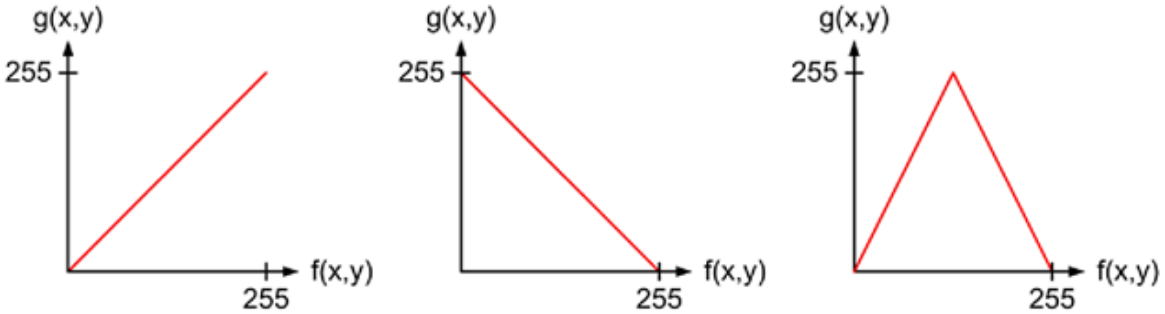
MATH:
LINEAR OPERATION

$$g(x,y) = \mathbf{a}\, f(x,y) + \mathbf{b}$$

f(x,y) : (i/p) image intensity @ (row x, column y)
g(x,y): (o/p) image intensity @ (row x, column y)
a : slope (contrast)
b : translation (shifting)

60	100	150	200
60	100	150	200
60	100	150	200
60	100	150	200

LINEAR OPERATION



(a) No effect (b) Invert (c) Solarize

THINGS TO BE CAREFUL

- What would be data type effects after performing math operation on pixel intensity?
  - Uint8 -> float
  - Out of range
    - Some pixels
      - **Values < 0** -> negative
      - **Values > 255** -> over range of uint8
  - Different Library Dealing data type conversion
    - Different way
      - Numpy
      - OpenCV

Numpy

MATH:
LINEAR OPERATION

$$g(x,y) = \mathbf{a}\, f(x,y) + \mathbf{b}$$

a = 2, b = 0

a = 1, b = -100

a = -1, b = 0

f(x,y)
uint8

60	100	150	200
60	100	150	200
60	100	150	200
60	100	150	200

g(x,y)
float64

g(x,y)
uint8

MATH:
LINEAR OPERATION

$$g(x,y) = \mathbf{a}\, f(x,y) + \mathbf{b}$$

a = 2, b = 0

a = 1, b = -100

a = -1, b = 0

f(x,y)
uint8

60	100	150	200
60	100	150	200
60	100	150	200
60	100	150	200

g(x,y)
float64

120	200	300	400
120	200	300	400
120	200	300	400
120	200	300	400

g(x,y)
uint8

120	200	44	144
120	200	44	144
120	200	44	144
120	200	44	144

MATH:
LINEAR OPERATION

$$g(x,y) = \mathbf{a}\, f(x,y) + \mathbf{b}$$

a = 2, b = 0

a = 1, b = -100

a = -1, b = 0

f(x,y)
uint8

60	100	150	200
60	100	150	200
60	100	150	200
60	100	150	200

g(x,y)
float64

g(x,y)
uint8

THINGS TO BE CAREFUL

OpenCV

```
w1 = 1
w2 = 1
img_add = cv2.addWeighted(Im_1,w1,Im_2,w2,0)

print(Im_1)
print(Im_2)
print(img_add)
```

```
[[ 60 100 150 200]
 [ 60 100 150 200]
 [ 60 100 150 200]
 [ 60 100 150 200]]
[[216   0  50 100]
 [216   0  50 100]
 [216   0  50 100]
 [216   0  50 100]]
[[255 100 200 255]
 [255 100 200 255]
 [255 100 200 255]
 [255 100 200 255]]
```

```
w1 = -1
w2 = 1
img_add = cv2.addWeighted(Im_1,w1,Im_2,w2,0)

print(Im_1)
print(Im_2)
print(img_add)
```

```
[[ 60 100 150 200]
 [ 60 100 150 200]
 [ 60 100 150 200]
 [ 60 100 150 200]]
[[216   0  50 100]
 [216   0  50 100]
 [216   0  50 100]
 [216   0  50 100]]
[[156   0   0   0]
 [156   0   0   0]
 [156   0   0   0]
 [156   0   0   0]]
```

THINGS TO BE CAREFUL

- Self conversion
  - **Perform Rescaling Intensity values**
    - **EX**
      - **Quantization**

120	130	140	150
120	130	140	150
120	130	140	150
120	130	140	150

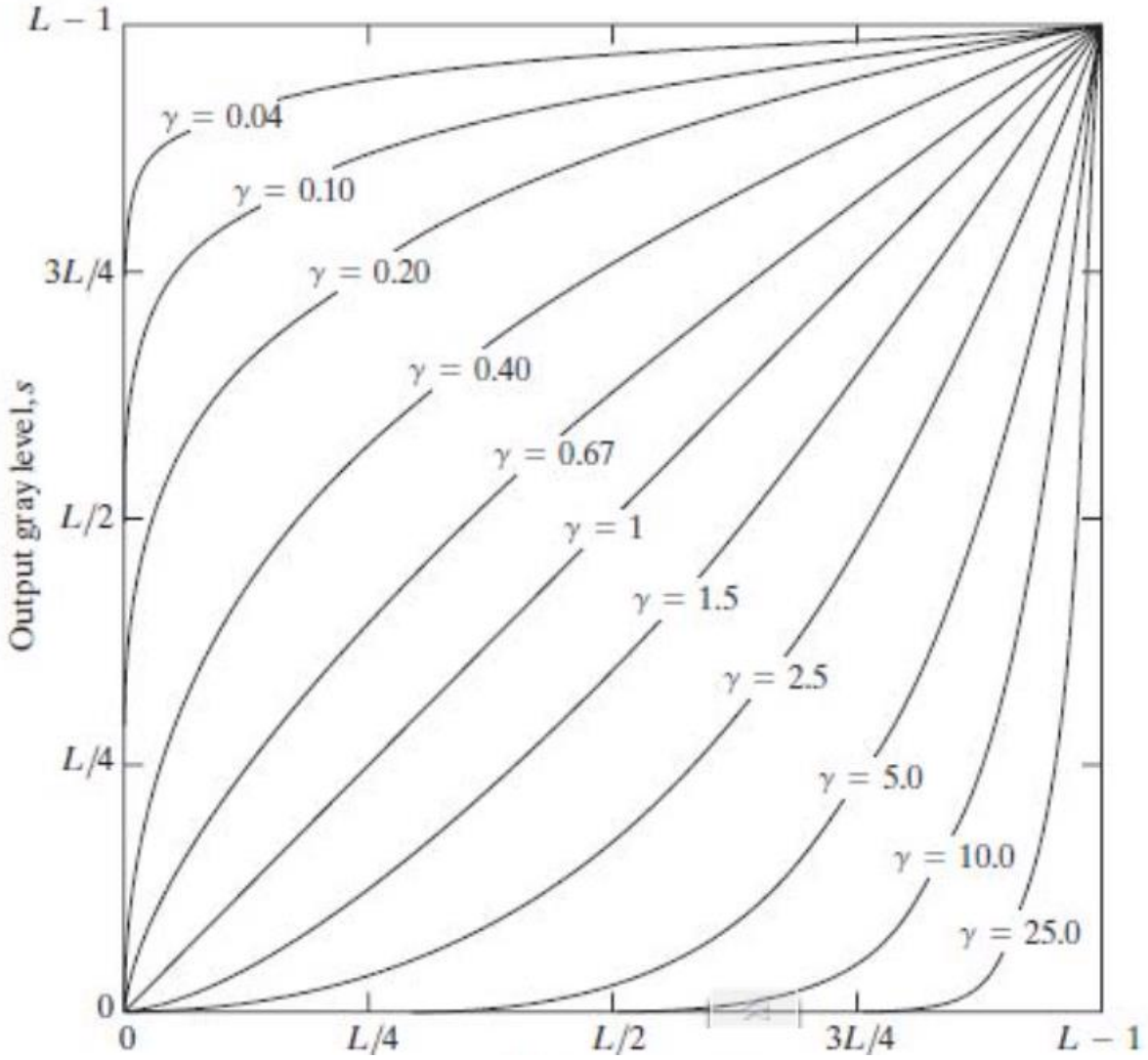
$$g(x,y) = \mathbf{a}\, f(x,y) + \mathbf{b}$$

max	-90
min	-135
max + ep	-89.9999
min - ep	-135.0001
bit_depth	8
Qlevel	256
(max-min) / Qlevel	0.1758

a =	-1.5	b =	90
-90	-105	-120	-135
-90	-105	-120	-135
-90	-105	-120	-135
-90	-105	-120	-135
255	170	85	0
255	170	85	0
255	170	85	0
255	170	85	0
ep	0.000100		

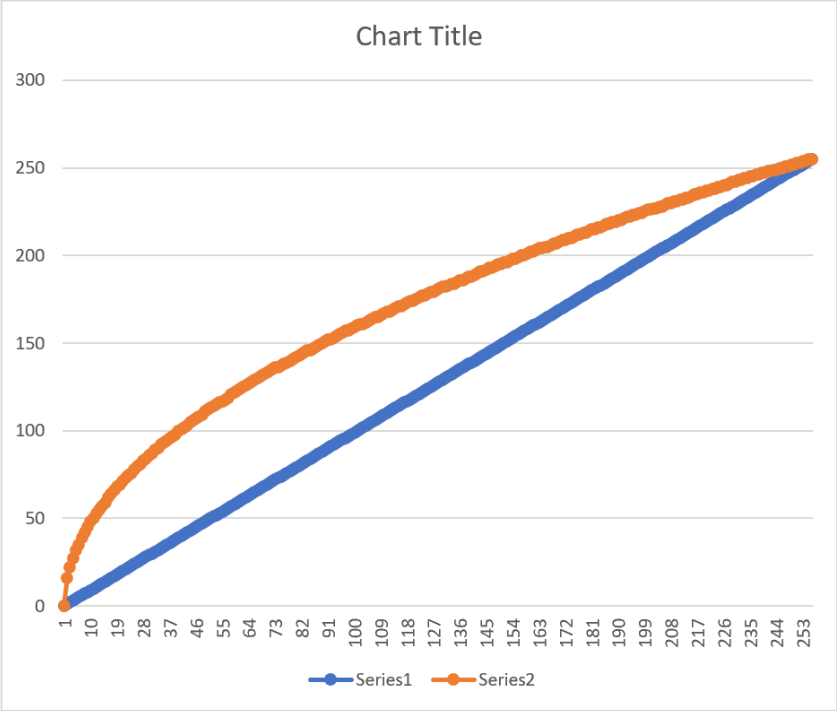
Math: Gamma Operation

- g(x,y) = a (f(x,y) ^ γ) + b
    - f(x,y) : (i/p) image intensity @ (row x, column y)
    - g(x,y): (o/p) image intensity @ (row x, column y)
    - a : slope (contrast)
    - b : translation (shifting)



Gamma Operation

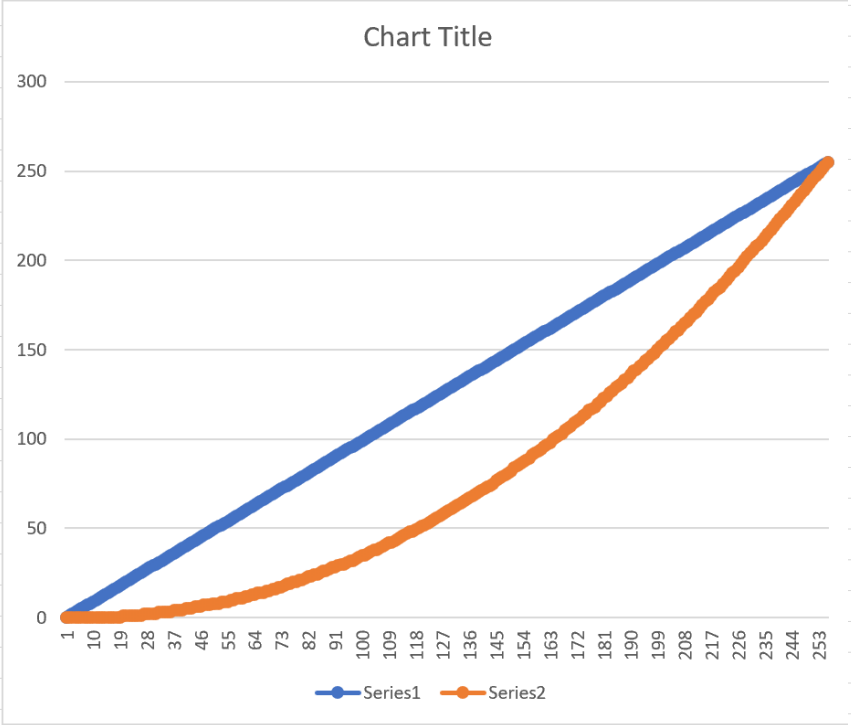
uint 8 Intensity	Linear			Gamma	
	y = a x +b			y = a x ^ gamma + b	
parameters	a	2		1	
	b	100		20	
	gamma			0.5	
	max	610.001		35.96972	
	min	100		20	
(max-min) / Qlevel		1.992191		0.062382	
			Quantize		Quantize
0		100	0	20	0
1		102	1	21	16
2		104	2	21.41421	22
3		106	3	21.73205	27
4		108	4	22	32
5		110	5	22.23607	35
6		112	6	22.44949	39
7		114	7	22.64575	42
8		116	8	22.82843	45
9		118	9	23	48
10		120	10	23.16228	50
11		122	11	23.31662	53
12		124	12	23.4641	55
13		126	13	23.60555	57
14		128	14	23.74166	59
15		130	15	23.87298	62
16		132	16	24	64
17		134	17	24.12311	66
18		136	18	24.24264	68
19		138	19	24.3589	69
20		140	20	24.47214	71
21		142	21	24.58258	73
22		144	22	24.69042	75
23		146	23	24.79583	76
24		148	24	24.89898	78
25		150	25	25	80



	Linear		Gamma
72	244	72	28.48528	136
73	246	73	28.544	136
74	248	74	28.60233	137
75	250	75	28.66025	138
76	252	76	28.7178	139
77	254	77	28.77496	140
78	256	78	28.83176	141
79	258	79	28.88819	142
80	260	80	28.94427	143
81	262	81	29	144
82	264	82	29.05539	145
83	266	83	29.11043	146
84	268	84	29.16515	146
85	270	85	29.21954	147
86	272	86	29.27362	148
87	274	87	29.32738	149
88	276	88	29.38083	150
89	278	89	29.43398	151
90	280	90	29.48683	152
91	282	91	29.53939	152
92	284	92	29.59166	153
93	286	93	29.64365	154
94	288	94	29.69536	155
95	290	95	29.74679	156

Gamma Operation

uint 8	**Linear**			**Gamma**	
Intensity					
	y = a x +b			y = a x ^ gamma + b	
parameters	a	2		1	
	b	100		20	
	gamma			2.1	
	max	610.001		113190.8	
	min	100		20	
(max-min) / Qlevel		1.992191		442.0734	
			Quantize		Quantize
0		100	0	20	0
1		102	1	21	0
2		104	2	24.28709	0
3		106	3	30.04511	0
4		108	4	38.37917	0
5		110	5	49.36547	0
6		112	6	63.06432	0
7		114	7	79.52589	0
8		116	8	98.79324	0
9		118	9	120.9042	0
10		120	10	145.8925	0
11		122	11	173.7888	0
12		124	12	204.6208	0
13		126	13	238.4143	0
14		128	14	275.1931	0
15		130	15	314.9794	0
16		132	16	357.794	0
17		134	17	403.6567	0
18		136	18	452.5858	0
19		138	19	504.5991	1
20		140	20	559.7131	1
21		142	21	617.944	1
22		144	22	679.3069	1
23		146	23	743.8165	1
24		148	24	811.4867	1

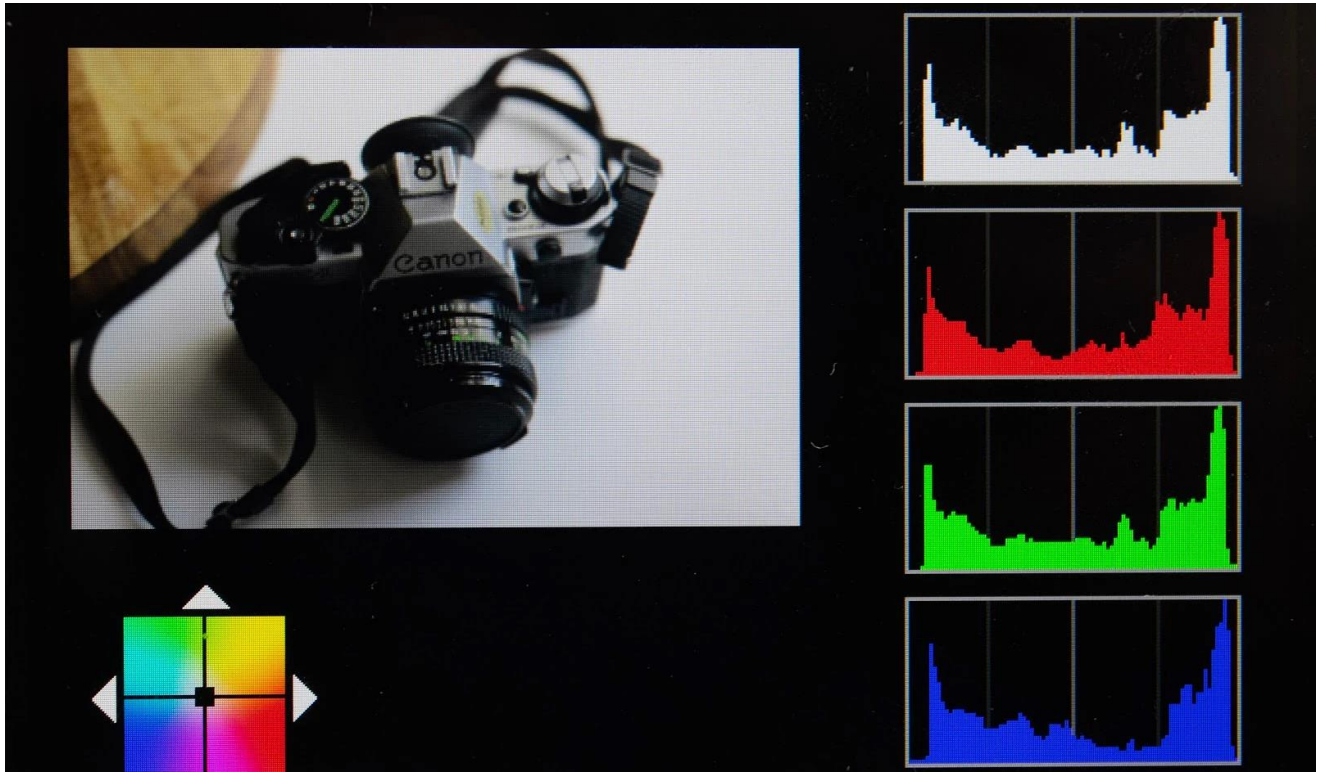


	Linear		**Gamma**	
71	242	71	7740.449	17
72	244	72	7970.57	17
73	246	73	8204.233	18
74	248	74	8441.445	19
75	250	75	8682.209	19
76	252	76	8926.53	20
77	254	77	9174.413	20
78	256	78	9425.863	21
79	258	79	9680.885	21
80	260	80	9939.482	22
81	262	81	10201.66	23
82	264	82	10467.42	23
83	266	83	10736.77	24
84	268	84	11009.72	24
85	270	85	11286.26	25
86	272	86	11566.4	26
87	274	87	11850.15	26
88	276	88	12137.52	27
89	278	89	12428.49	28
90	280	90	12723.09	28
91	282	91	13021.3	29
92	284	92	13323.15	30
93	286	93	13628.62	30
94	288	94	13937.73	31
95	290	95	14250.48	32
96	292	96	14566.87	32
97	294	97	14886.91	33
98	296	98	15210.59	34
99	298	99	15537.93	35
100	300	100	15868.93	35
101	302	101	16203.59	36
102	304	102	16541.91	37
103	306	103	16883.91	38

STATISTICAL BASED OPERATION

Adjust Pixel Intensity

Statistical based operation



- What type of statistics can we use to
  - Adjust pixel intensity?

- Frequency statistics
  - Histogram
    - Count #pixels of each
      - Pixel intensity

    - Normalized histogram
      - Probability of occurrence
      - Percent of each pixel intensity

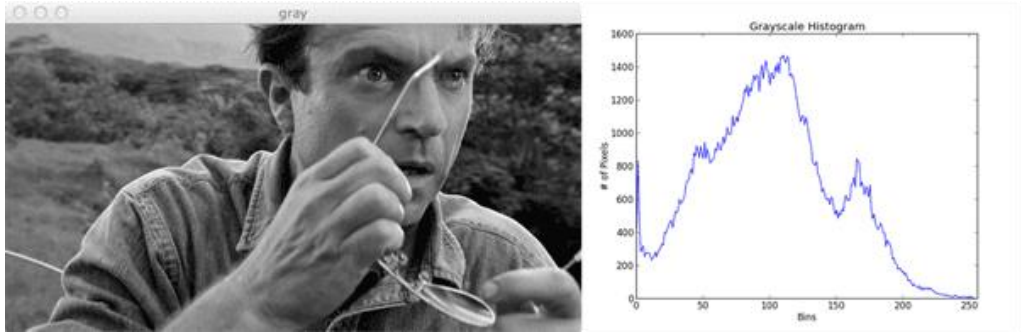
Statistical based operation

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]

What can Histogram tell us?



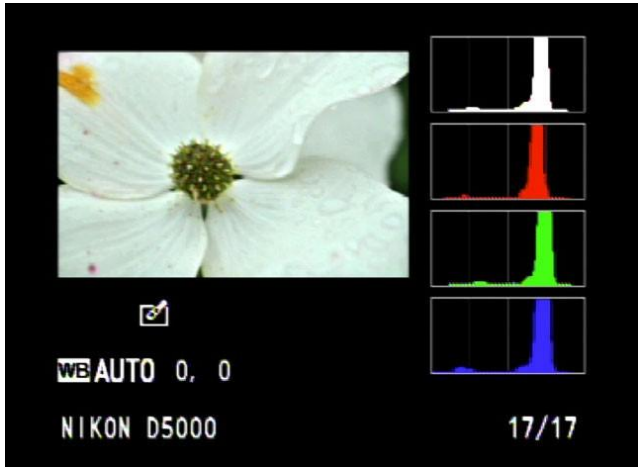
Statistical distribution of

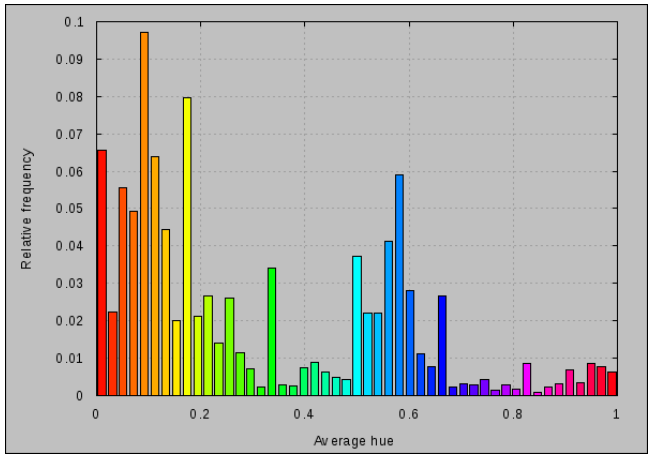
Grayscale image

- Brightness distribution
- Single histogram

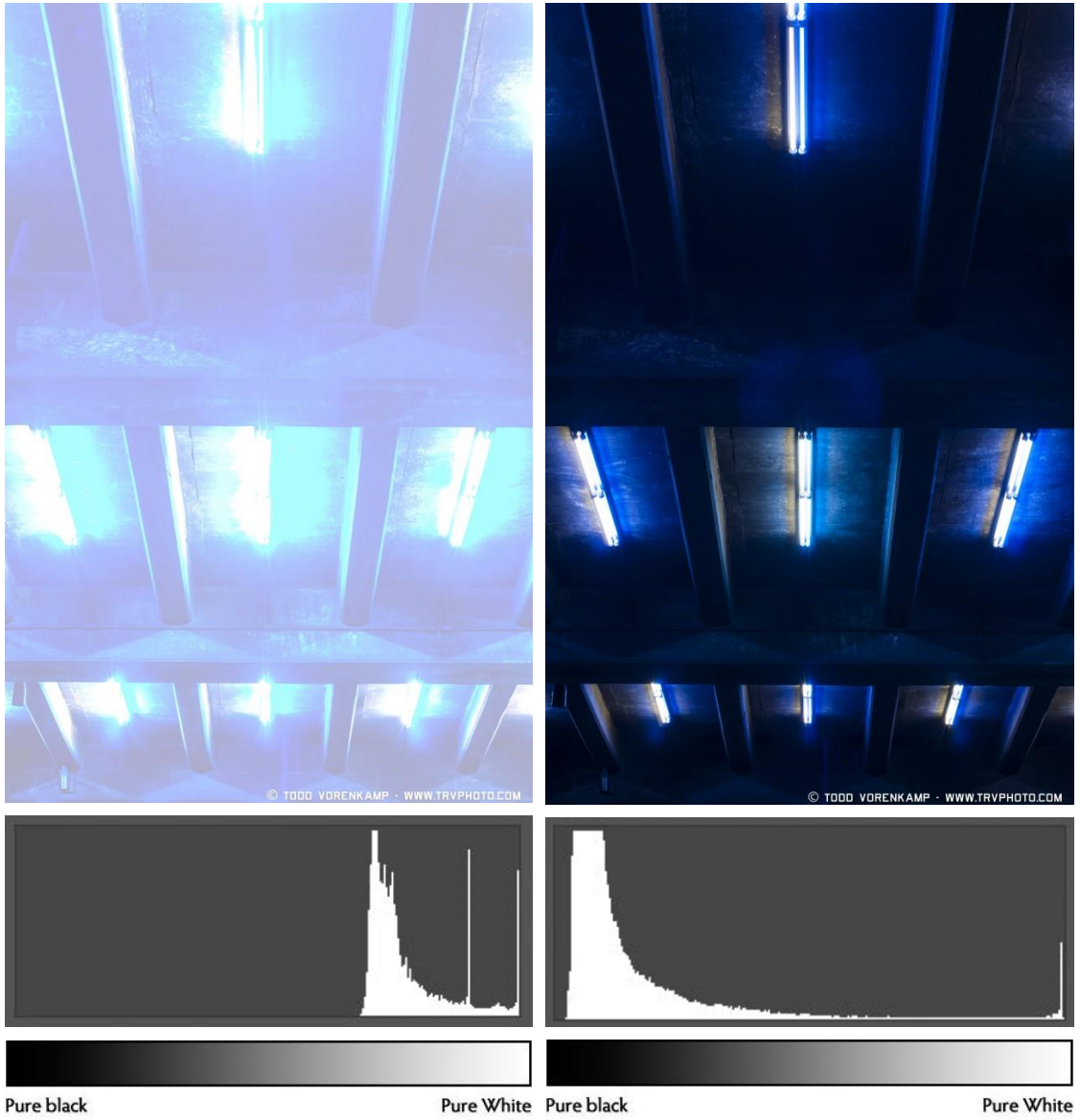
Color image

- Color distribution
- Separated histogram
- Combined histogram





Can histogram tell us image quality?

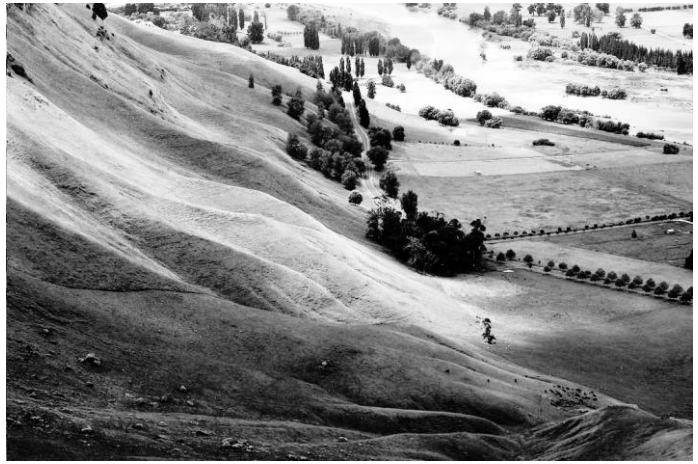


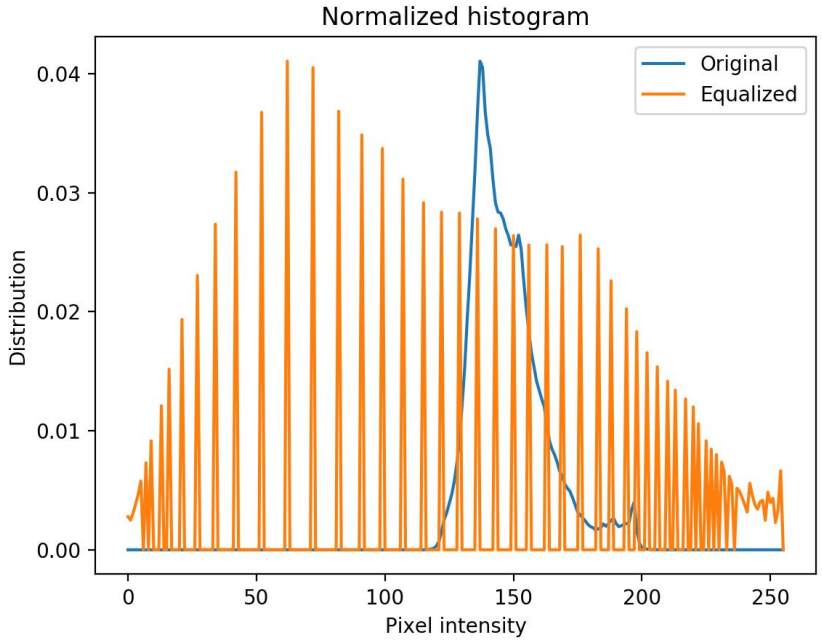
HISTOGRAM EQUALIZATION

Adjust Pixel Intensity

Histogram Equalization

- กระจาย เฉดสี ให้เท่าๆกัน
  - histogram output
    - พยายามทำให้ผลลัพธ์ใกล้เคียง uniform distribution
  - สำหรับทำ
    - Contrast enhancement

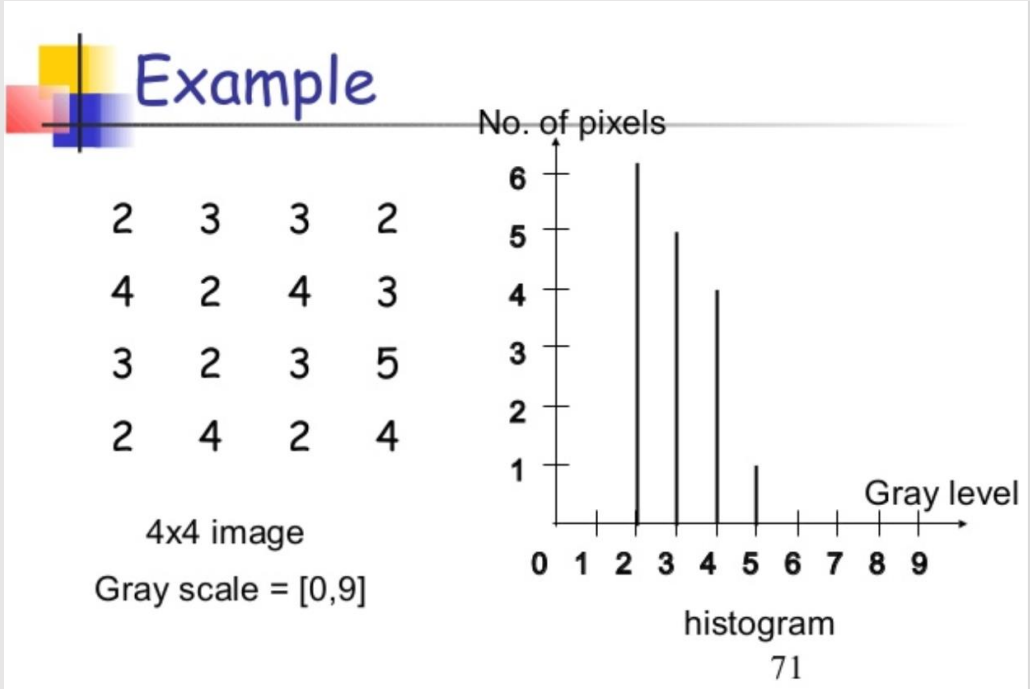




https://levelup.gitconnected.com/introduction-to-histogram-equalization-for-digital-image-enhancement-420696db9e43

Math:
Histogram equalization Operation

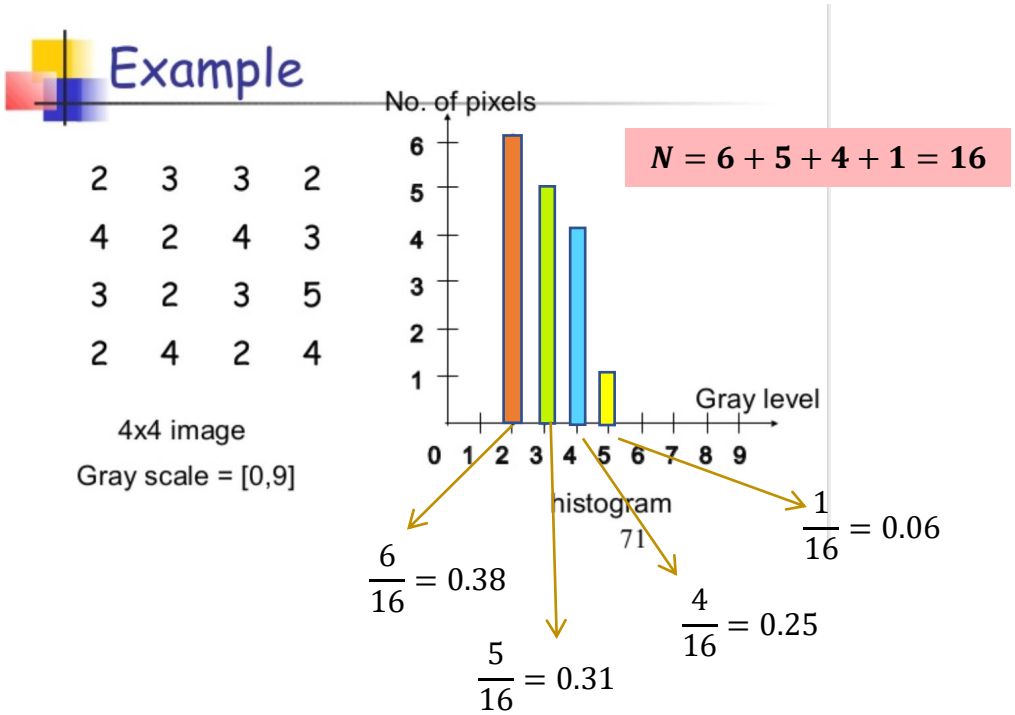
Intensity level	0	1	2	3	4	5	6	7	8	9
Hist	0	0	6	5						
Normalized	0	0	6/16	5/16						
CDF	0	0	6/ 16	11/ 16						
Equalized intensity	0	0	(6/16) * 9 =3.33 =3							

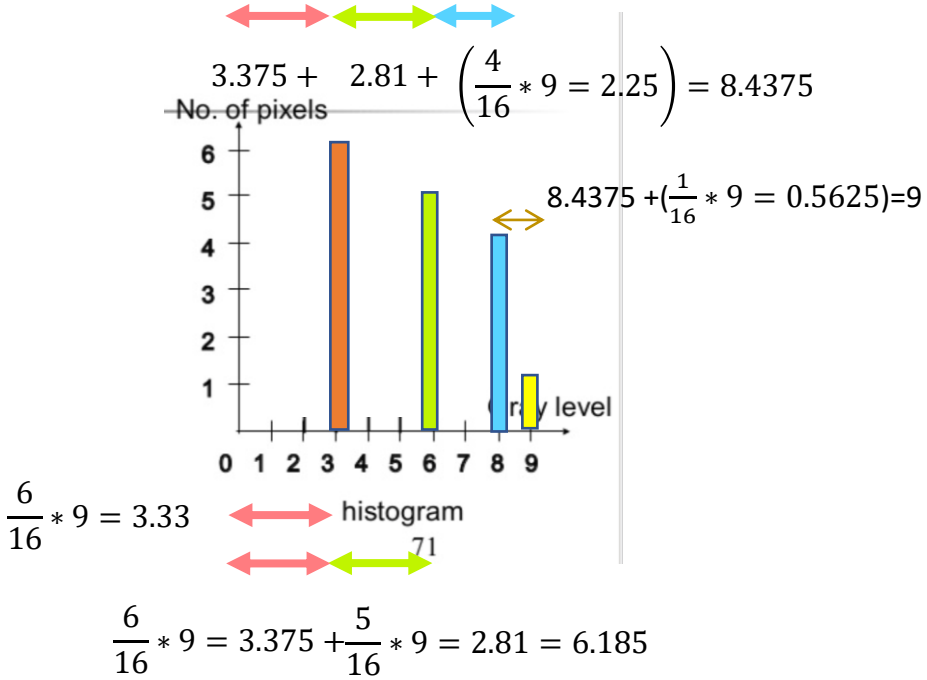


CDF: Cumulative Density Function (Cumulative Histogram))

Histogram Equalization

- Results
  - กระจายเฉดแสงให้ห่างกัน ตามสัดส่วนเชิงปริมาณ (ความน่าจะเป็น) ของเฉดสีนั้นๆ
    - ถ้าเฉดสี มี สัดส่วน หรือ ความน่าจะเป็น มาก
      - จะให้ เปลี่ยนเฉดสีนั้น ให้มี ระยะความห่างของเฉดสีนั้น กับ เฉดสีก่อนหน้า ด้วยระยะห่างมาก
        - เพื่อให้เกิด contrast หรือความแตกต่างของเฉดสีที่อยู่ใกล้กัน ให้ห่างกันหรือต่างกันมากขึ้น





Histogram Equalization

- Results
  - กระจายเฉดแสงให้ห่างกัน ตามสัดส่วนเชิงปริมาณ (ความน่าจะเป็น) ของเฉดสีนั้นๆ
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        - เพื่อให้เกิด contrast หรือความแตกต่างของเฉดสีที่อยู่ใกล้กัน ให้ห่างกันหรือต่างกันมากขึ้น

After pixel intensity transform

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

Example

3	6	6	3
8	3	8	6
6	3	6	9
3	8	3	8

4x4 image

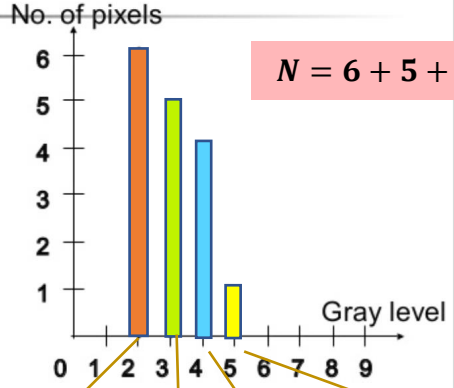
Gray scale = [0,9]

Example

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

4x4 image

Gray scale = [0,9]



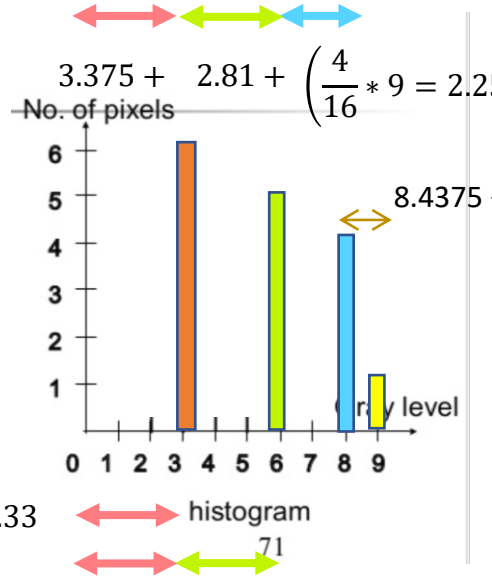
$N = 6 + 5 + 4 + 1 = 16$

$\frac{6}{16} = 0.38$

$\frac{5}{16} = 0.31$

$\frac{4}{16} = 0.25$

$\frac{1}{16} = 0.06$



$3.375 + \quad 2.81 + \left(\frac{4}{16} * 9 = 2.25\right) = 8.4375$

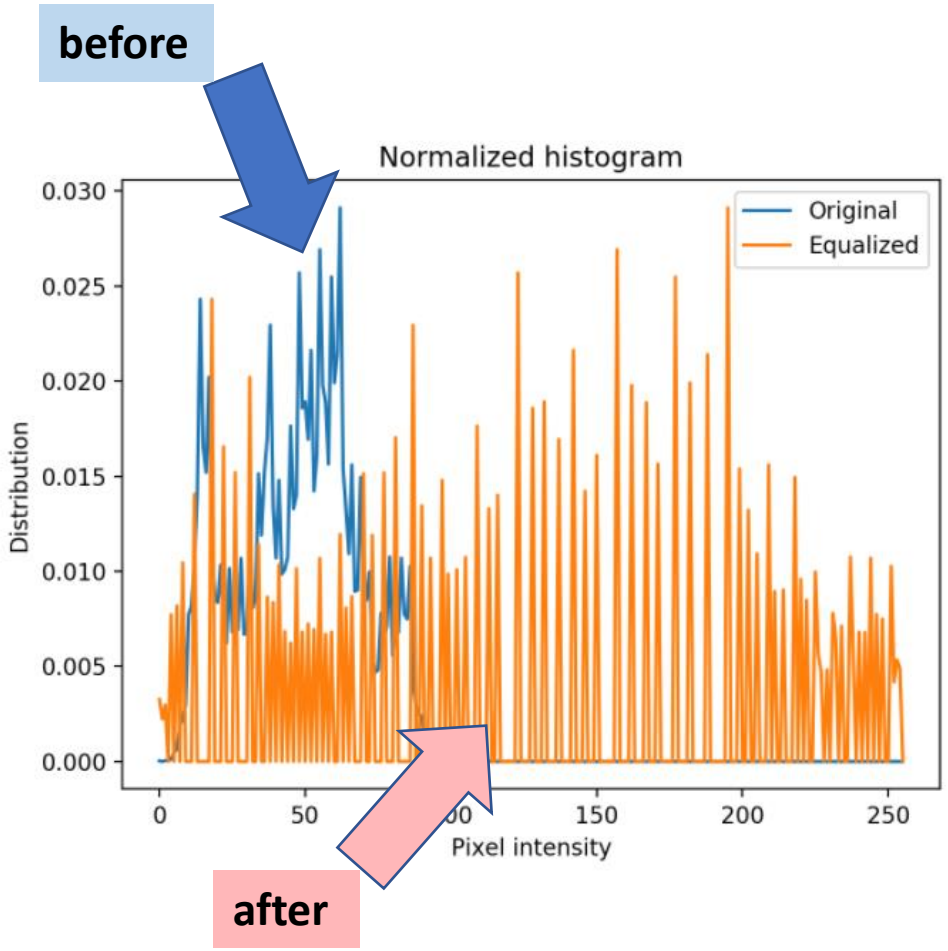
$8.4375 + (\frac{1}{16} * 9 = 0.5625) = 9$

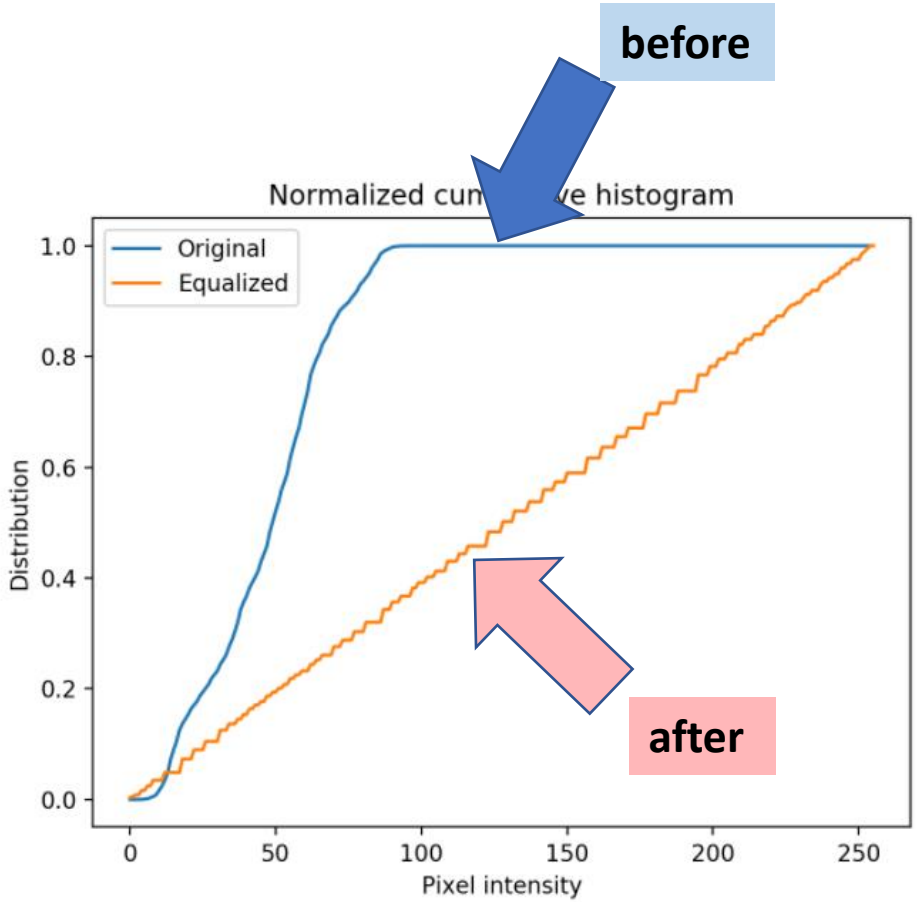
$\frac{6}{16} * 9 = 3.33$

$\frac{6}{16} * 9 = 3.375 + \frac{5}{16} * 9 = 2.81 = 6.185$

Normalized Histogram / Cumulative Histogram

(before and after performing histogram equalization)





https://levelup.gitconnected.com/introduction-to-histogram-equalization-for-digital-image-enhancement-420696db9e43

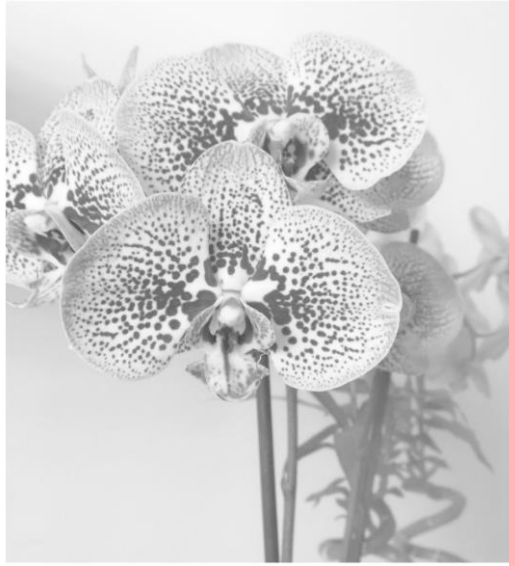
HISTOGRAM MATCHING

Adjust Pixel Intensity

Histogram matching

- ปรับเทียบ กระจาย เฉดสี
  - จาก Histogram ภาพต้นฉบับ
    - Hist (input image)
  - ไปตาม Histogram ต้นแบบ ที่ต้องการใดๆ
    - Hist (template image)
  - histogram output
    - พยายามทำให้ผลลัพธ์มีการการจายเฉดสีใกล้เคียงกับ histogram ต้นแบบ
  - สำหรับทำ
    - Contrast enhancement

Input image



Template image

Output image

Histogram Matching (Calculation)

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

input: Image A

Pixel Value	Histogram	Equalized Histogram
0	4	4
1	5	9
2	3	12
3	6	18
4	7	25

Equalized Histogram of A

Pixel value	Normalized Hist		Cumulative Normalized Hist	
0	4/25 =	0.16	4/25 =	0.16
1	5/25 =	0.2	9/25 =	0.36
2	3/25 =	0.12	12/25 =	0.48
3	6/25 =	0.24	18/25 =	0.72
4	7/25 =	0.28	25/25 = 1	1

Input image

2	1	2	1	0
3	3	2	4	4
1	3	2	4	4
0	0	3	2	1
1	3	1	4	0

Target: Image B

Pixel Value	Histogram	Equalized Histogram
0	4	4
1	6	10
2	5	15
3	5	20
4	5	25

Equalized Histogram of B

Pixel value	Normalized Hist		Cumulative Normalized Hist	
0	4/25 =	0.16	4/25 =	0.16
1	6/25 =	0.24	10/25 =	0.4
2	5/25 =	0.2	15/25 =	0.6
3	5/25 =	0.2	20/25 =	0.8
4	5/25 =	0.2	25/25 = 1	1

Template image

https://towardsdatascience.com/histogram-matching-ee3a67b4cbc1

Histogram Matching (Calculation)

Input image

0	2	1	3	4
1	3	4	3	3
0	1	3	1	4
3	1	4	2	0
0	4	2	4	4

input: Image A

Pixel value	Normalized Hist		Cumulative Normalized Hist	
0	4/25 =	0.16	4/25 =	0.16
1	5/25 =	0.2	9/25 =	0.36
2	3/25 =	0.12	12/25 =	0.48
3	6/25 =	0.24	18/25 =	0.72
4	7/25 =	0.28	25/25 = 1	1

Template image

2	1	2	1	0
3	3	2	4	4
1	3	2	4	4
0	0	3	2	1
1	3	1	4	0

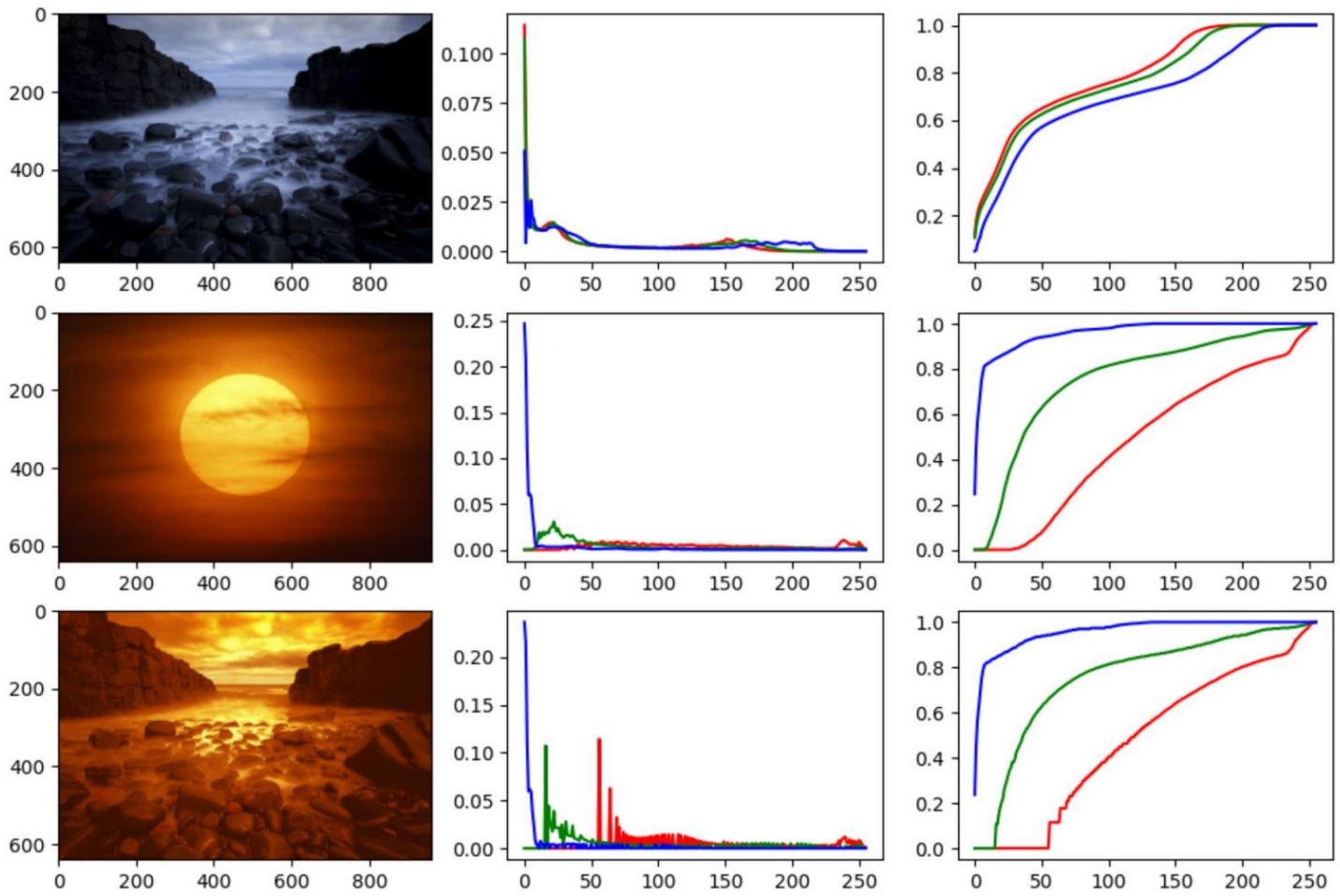
Target: Image B

Pixel value	Normalized Hist		Cumulative Normalized Hist	
0	4/25 =	0.16	4/25 =	0.16
1	6/25 =	0.24	10/25 =	0.4
2	5/25 =	0.2	15/25 =	0.6
3	5/25 =	0.2	20/25 =	0.8
4	5/25 =	0.2	25/25 = 1	1

Input Image	Cumulative Normalized Hist (input)		Cumulative Normalized Hist (Tempmlate)		Template Image	Mapping output
0	(4/25)*4 =	0.16	4/25 =	0.16	0	0
1	(9/25)*4 =	0.36	10/25 =	0.4	1	1
2	(12/25)*4 =	0.48	15/25 =	0.6	2	1
3	(18/25)*4 =	0.72	20/25 =	0.8	3	3
4	(25/25)*4 = 1	1	25/25 = 1	1	4	4

https://towardsdatascience.com/histogram-matching-ee3a67b4cbc1

Histogram Matching (Calculation)



CDF input

CDF template

CDF output

Histogram based Image Enhancement

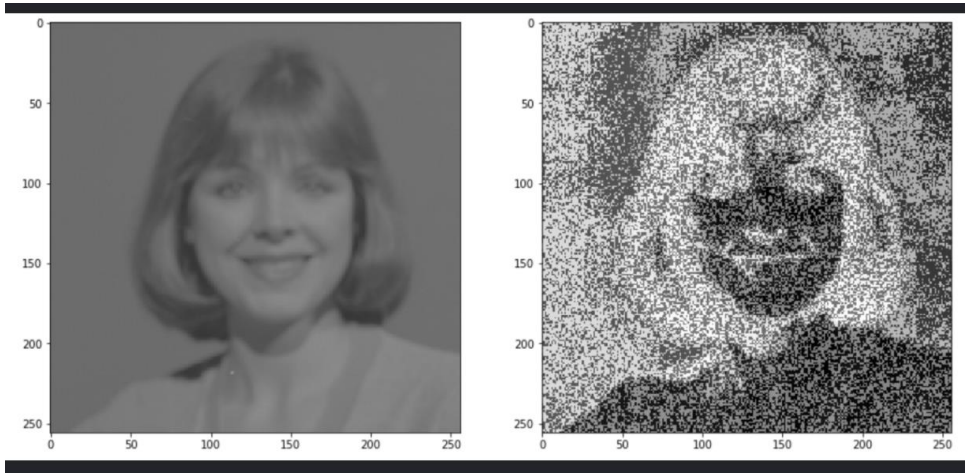
Advantage	Disadvantage

• Automatic operation
  • Using self image statistics
 	
• ลดภาระในการปรับ tune parameter
	• Unsuitable Statistics
	  • may lead to worse results

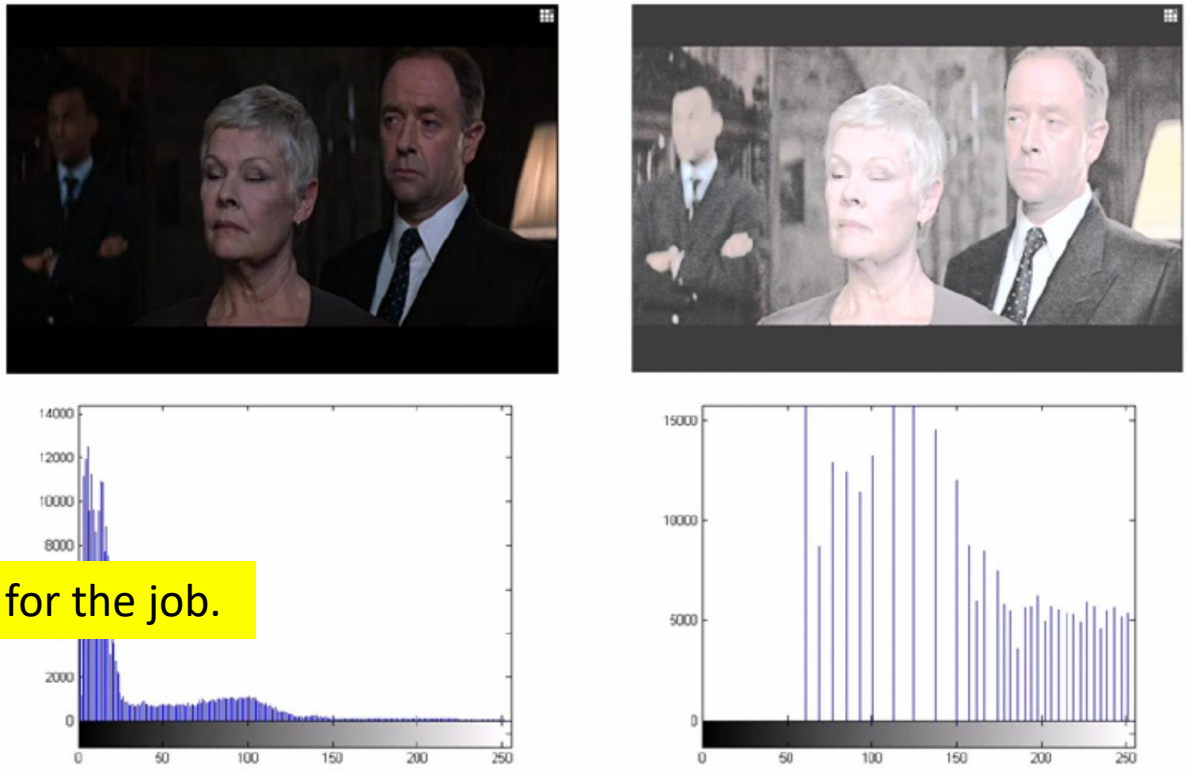
WILL THE STATISTICS BE GOOD FOR ALL IMAGES?

Adjust Pixel Intensity

Unsuitable image statistics for histogram equalization



https://hackernoon.com/histogram-equalization-in-python-from-scratch-ebb9c8aa3f23



Histogram equalization isn't always the perfect tool for the job.

https://www.researchgate.net/figure/Effect-of-Histogram-Equalization-The-image-on-the-top-left-has-a-histogram-with-a-large_fig1_229036250

TODAY'S TOPICS

- Image Enhancement
  - Pixel Intensity Adjustment
    - Math Operation
      - Linear / Nonlinear (gamma) function
    - Statistics based Operation
      - Histogram Equalization
      - Histogram Matching